CHE 435/525 Process Systems Analysis and Control (Spring 2025)

Instruction Team

Instructor **Prof. Wentao Tang**
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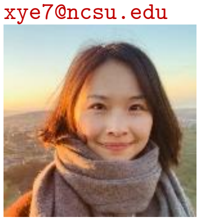

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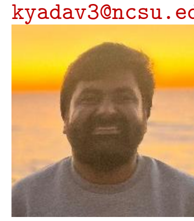

Time and Location

Lectures: Tue 8:30am – 9:45am & Thu 8:30am – 9:45am, *EB1 1011*

- Class size (counted at the start of the semester): **130**
- Section 002 (*in classroom*): 107 (CHE 435) + 10 (CHE 525)
- Section 001 (*asynchronous*): 13 (CHE 435)
  - Recorded lectures available in this Panapto folder.

Exams:

- **Midterm Exams:** 2/19 Wed & 4/2 Wed 6:30pm – 8:00pm, *EB2 1025* (surnames A—M) / *EB2 1231* (surnames N—Z)
- **Final Exam:** 4/24 Thu 8:30 – 11:00am, ***Hunt 1103*** (Section 002) / 4/30 Wed 8:30 – 11:00am, *Fitts 2331* (Section 001)

Office Hour Schedule

	Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
1:00—2:30pm	**Addington**			**Addington**	
2:30—4:00pm	**Yadav**	**Wang**	**Yadav**	**Fasiku**	**Wang**
4:00—5:30pm		**Tang**	**Tang**	**Fasiku**	**Ye**

Course Prerequisites

Formal prerequisites: C- or better grades in (MA 341 and TE 205) or CHE 312.
Detailed technical prerequisites: Calculation with complex numbers, simple factorization of polynomials, solution of quadratic algebraic equations, calculation with fractional expressions, conversion of fractional equations into polynomial equations, integration and differentiation, analytical solution of ODEs. Basic MATLAB (or other programming language) for numerical computation. Motivation to learn, even in the final semester.

Course Objectives

By the end of the course, you should be able to: (1) Develop a linearized process model from first principles and solve using methods such as MATLAB and/or Laplace transforms. (2) Design and tune a feedback control system utilizing PID control for a given process model and analyze its stability. (3) Develop effective model-based control systems with feedforward control, inferential control, and dead-time compensation. (4) Incorporate hazard mitigation in the design of a process control scheme, including provisions for emergency relief and shutdown.

Textbooks

Primary Textbook:

Either D. E. Seborg, et al., Process dynamics and control, Wiley, 4th ed., 2016,

This book is technically *less difficult* (but sometimes oversimplified) and more widely used in universities. The instructor adopts this textbook only because it is considered to be easier for the students.

or C. Kravaris, I. K. Kookos. *Understanding process dynamics and control*, Cambridge University Press, 2021.

This book is more *rigorous and logical*, modern, but maybe more challenging. It is suitable for those who want to have a solid grasp of fundamental principles, prepared in mathematics, and/or will do control in the future.

- In the tentative course schedule below, the correspondence between the lectures and Seborg's chapters is provided. Students who use the other book should find the corresponding chapters.
- Some lecture contents, unfortunately, are covered by only one of the two books. This is due to the richness of today's control practice and computing technology. The instructor feel necessary to expose students to such "advanced" topics but with lesser requirement. The lecture notes should suffice.
- When the contents written on the first textbook are inconsistent with the lectures, use the approach taught in the lectures. This can happen due to it being an older textbook.
- Most homework problems will be from the first book. Some homework problems are from the second book.

Recommended supplemental texts:

- For a more *comprehensive* introduction, especially if your future work will be in chemical engineering and relevant to process control, or if you would like to understand process control just a bit more thoroughly, read: **B. A. Ogunnaike, W. H. Ray, *Process dynamics, modeling, and control*, Oxford University Press, 1994.**
- For a more *thorough and detailed* methodology in controller design and implementation, especially if you are interested in applications beyond chemical processes (such as mechatronic systems), or if you will do a technical job in control, read: **R. C. Dorf, R. H. Bishop. *Modern control systems*, 14th ed., Pearson, 2022.**
- For more *advanced* methods that address the real-world control problems including multi-input-multi-output systems, especially if you will become a "person in control" for a long-term career, read (now or in the future): **S. Skogestad, I. Postlethwaite, *Multivariable feedback control: Design & analysis*, Wiley, 2nd ed., 2005.**
- If you are interested in knowing about today's research frontiers of process control or doing research in this area, you are always encouraged to discuss with the instructor.

Calculation of Course Grades

A weighted average score will be calculated as follows:

- Mid-Term Examinations (2) = **39%** (26% for the higher, 13% for the lower)
- Final Examination (1) = **26%**
- Regular homework problems (7 sets) = **35%** (5.0% for each homework set)
- Advanced homework problems (7 sets) = Bonus **7%** (1.0% for each homework set)
- Teaching surveys = Bonus **3%** (1.5% for mid-term survey, 1.5% for ClassEval)
    - For students taking the course under CHE 525, advanced homework problems are *required* instead of optional for bonus. Regular homework problems and advanced problems altogether count as 35%. Bonus points for doing teaching surveys do not apply.

The final letter grade is assigned according to the score *per se* or according to the *quantile* of the score in the class. The threshold values are listed in the table below.

- *Example*: If your weighted average score is 82 (B+), and that is higher than or equal to 87% of the class (A), then your letter grade is **max**(B+, A) = A.
- The class size of CHE 435 *after the official drop/revision deadline* is used to calculate the quantile.
- For students taking the course under CHE 525, the quantile criterion does not apply.

Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Score	95	90	85	80	75	70	65	60	55	50	45	40	0
Quantile	95%	85%	75%	65%	55%	45%	35%	25%	15%	10%	5%	0%	0%

Tentative Course Schedule

Lec.	Date	Subject	Read (Seborg)	Due	Part
1	1/7	Motivation, Illustrative example	§1.1—1.5	HW1 [1/17, Fri]	*I. Process Modeling*
2	1/9	Tank models, Degree-of-Freedom analysis	§2.1—2.4		
3	1/14	Reactor models, Simulation of nonlinear systems	§2.4—2.5		
4	1/16	Laplace transform, Solution of ODE, PFE	§3.1—3.3	HW2 [1/31, Fri]	*II. Dynamical Analysis*
5	1/21	Laplace transform properties, Qualitative behavior	§3.4—3.6		
6	1/23	Transfer function, Linearization of nonlinearity	§4.1—4.3		
7	1/28	First-order systems, Integrating systems	§5.1—5.3		
8	1/30	Second-order systems, Damping, Oscillations	§5.4	HW3 [2/14, Fri]	
9	2/4	High-order system, Poles, Stability, Delayed system	§6.2—6.4		
10	2/6	Zeros, Inverse response, Invertibility	§6.1, Notes		
		Wellness Day (2/11)		HW4 [2/28, Fri]	
11	2/13	Frequency response, Bode diagrams	§14.1—14.2		
12	2/18	**Further examples** of Bode diagrams, Sketching	§14.3, Notes		
E1	2/19	**Exam 1** to Lec. 10 (incl. dynamics except for frequency domain)			
13	2/25	Feedback loop, PID controllers, Block diagram	§8.1, 8.2, 11.1	HW5 [3/17, Mon]	*III. Classical Process Control*
14	2/27	Closed-loop transfer function, Offset, Stability	§11.2, 11.3, 8.5		
15	3/4	Closed-loop stability, Performance metric, Simulation	§11.4, 12.1		
16	3/6	Direct controller synthesis, Internal model control	§12.2, Notes	HW6 [3/31, Mon]	
		Spring Break (3/10 — 3/14)			
17	3/18	**Further examples** of controller synthesis	Notes		
18	3/20	Feedforward-feedback control loop, synthesis	§15.4, Notes		
G1	3/25	*Guest Lecture:* Introduction to Multivariable Systems [Dr. Xiuzhen Ye, NC State Univ.]	Guest Slides		
19	3/27	Bode diagram of the loop, Bode stability criterion	§14.4—14.6	HW7 [4/14, Mon]	
20	4/1	Stability margins, Robust stability, **Examples**	§14.7, Notes		
E2	4/2	**Exam 2** to Lec. 18 (incl. control except for frequency domain)			
21	4/8	Compensator design, PI & lead-lag design, **Examples**	Notes		
22	4/10	Cascading, Inferential control, Pairing and decoupling	§16.1, 16.3, 18.2		
23	4/15	Real-time optimization, Linear & quadratic program	§19.1—19.4	No HW	*IV. "Advanced" Process Control*
G2	4/17	*Guest Lecture or Lecture:* Model Predictive Control [Dr. Xue Yang, Torc Robotics]	Guest Slides		
G3	4/22	*Guest Lecture:* Multiparametric MPC & Process Intensification [Prof. Yuhe Tian, West Virginia Univ.]	Guest Slides		
EF		**Final Exam** to Lec. 22 (incl. frequency domain)			

Notes

1. All homework submissions are due at **11:59pm** of the stated deadline.
2. Office hours on the Wellness Day and its two adjacent days and the week of Spring Break are cancelled.
3. The schedule is tentative and subject to changes throughout the semester if necessary.

POLICIES AND PROCEDURES

- **Attendance:** Although the instructor does not check attendance, attendance of the lectures is expected. Documented excuses for being absent should be presented to the instructor. For a full statement of the university attendance policy, see here.
  - Except for medical reasons or emergent situations, most of the commonly seen reasons for being absent from classes are unjustified. These pretexts include absurd ones (e.g., alarm did not work) as well as seemingly "decent" ones (such as but not limited to, visiting a graduate school, having job interviews, visiting families over extended durations, working with classmates on another course). You should be able to reschedule these activities or ask for accommodation.
  - It is taken for granted that one who attends a university, chooses a major, and registers for a course should be committed to their study. A student who is absent from classes without prior consent or justifiable reasons is considered by the instructor to have degraded work ethics. You are then fully responsible for possible outcomes, such as the decline in academic performance.

- **Academic integrity.** Students should refer to the University policy on academic integrity in the Code of Student Conduct.
  - It is the instructor's understanding that the student's signature on any test or assignment means that the student contributed to the assignment in question and that they neither gave nor received unauthorized aid. Authorized aid on an individual assignment includes discussing the interpretation of the problem statement, sharing ideas or approaches for solving the problem, and explaining concepts involved in the problem. Any other aid would be unauthorized and a violation of the academic integrity policy. Unauthorized aid additionally includes accessing on-line solutions, whether that be posted copies of textbook solution manuals, "assistance" websites, AI-based technology, solutions files from any source, or paying someone else for homework writing. The use of any solution posted anywhere online is considered cheating and strictly forbidden. Any computer work must be completed on your personal computer or from your own account.
  - During exams, students are not allowed to use any means of electronic communication. Any use of a "connected" device without specific authorization during the exam will lead to cancellation of the exam paper without the right of retaking the exam. All cases of cheating will be reported immediately to the NCSU disciplinary committee, and the students will be removed from the class.
  - All documents – lecture notes, lecture videos, homework assignments and solutions, exams and solutions, or handouts – made available for this course are intended only for your personal use. **You are not allowed to share any content of the class** with any person not signed up for the course this semester; a personal, public, or commercial website; or any other news or advertising media.
  - All cases of academic misconduct will be submitted to the Office of Student Conduct. Students found guilty of academic misconduct will be subject to, at a minimum, a zero on the assignment in question, up to a failing grade in the course, depending on the nature of the violation. Students found guilty of academic misconduct in the course will be on academic integrity probation for the remainder of years at NCSU and may be required to report the violation on future professional school applications.

- **Homework.** Students should submit homework individually based on their *original works*.
  - You will use **Gradescope** to submit homework online. *Keep the hard copy until your grade is posted.*
  - Write on white, *unlined paper*, on *one side* of each page. *Begin each problem on a new page. Box all final numerical answers*. Write legibly and dark enough, and make sure the scan is clear. Use Scannable or another scanning app as opposed to taking photos. The problems must be submitted in the same order as in the assignment. In order to encourage you to follow these instructions, standard point deductions (-5 per error above) will be assigned for each violation.
  - **For computer problems**, the original codes, figures, and tables should be submitted along with the hand-written formal answers. You are recommended to use Matlab, Python, or Julia. Codes written in Matlab should be exported as a PDF file using the "**Publish**" option. Codes in Python or Julia should be created using Jupyter Notebook, Google Colab, or similar editors, and exported or printed in PDF format. Figures must be readable and the line series are properly positioned. Wherever applicable, axes should have **coordinates, labels, and units**. Different line series should use different colors, markers, or line types and should have **legends**.
  - Assignments should be submitted by the stated deadline. **All requests on homework deadline extensions will be rejected.** Late assignments will receive a 20-point deduction. Late homework will be accepted until

11:59pm 2 days after the due date. Once an individual hands in two late assignments, late homework will no longer be accepted.

- **Asking questions via email**. We strongly encourage you to discuss academic or personal questions with the instructor and TAs in office hours. But e-mails on homework problems are discouraged.
  - *If you must ask homework problems via emails, they should be addressed to the TAs only. The instructor and TAs may not answer your questions out of the office hours.*
  - *You must describe your* ***concrete and specific*** *questions clearly, and all your questions must be presented at once. The instructor and TAs will ask you to re-state the questions or refuse to answer when the questions are unclear.* (Example: "My calculation of this question does not appear correct. What should I do?" In this case, show all your work with numbered formulas, so that the TA may spot and point out the errors, instead of supposedly doing a full calculation for you.)
  - *If formulas or figures are involved, present them in a neat and readable way. You cannot expect the instructor or TAs to use formulas or draw figures in a brief email response.*
  - *Do not go back and forth. The instructor and TAs can refuse to respond if the discussions become tedious with excessive rounds of questions.*

- **Posted solutions.** Solutions answers to each problem will be posted on Moodle. However, *the solution process can be concise or omit intermediate steps*. It is your responsibility to make sure you find out how to solve the problems by asking about them in class or during office hours. You may also consult classmates to compare solutions after the graded homework has been returned.

- **Exams**. There will be two (2) exams during the semester and one (1) comprehensive final exam. The mid-term tests are 90 minutes long. The final exam is 150 minutes.
  - **All tests will be closed-book, closed-notes**. A practice test will be provided for each exam 10 days beforehand, which is expected to be slightly more difficult than the actual exam. For each exam, you are allowed to bring unlimited number of "**torpedo sheets**" hand-written on both sides. You may not include homework problems or worked-out examples on the torpedo sheet (the penalty is –10 points on the exam score if violated). You are only allowed to use your own sheet (making a copy of someone else's is an *academic integrity violation*!), and sign on all written sides. All torpedo sheets must be submitted with your completed exam and will not be returned – you can make backup copies before the exam if you want to reuse them for later exams.
  - **Missed tests**. Students who know in advance that they are unable to take a test during the common time must provide a documented excuse by the **Friday** of the week prior to the week of the test to be offered an alternate time for exam. Documented excuse provided after the aforementioned time are not accepted. Makeup tests will be held at a designated time during the last week of the semester.

- **Test and homework regrading**. If you believe that an error was made in grading, you can submit a re-grade request through Gradescope, no later than **five days** after the homework or test is returned. You need to be specific in your rationale of why you believe the problem was graded incorrectly. The grader or one of the instructors will review your request and respond. Any exam submitted for regrading will be entirely regraded to ensure fairness and accuracy.

- **Instructor's commitment.** You can expect your instructor to be courteous, punctual, well organized, and prepared for lecture and other class activities; to answer *questions* clearly and in a nonnegative fashion as long as these questions are reasonable and well-phrased; to be available during office hours or to notify you beforehand if unable to keep them; to provide a suitable guest lecturer when traveling; and to grade uniformly and consistently according to the posted guidelines.

- **Students with disabilities.** Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (REG02.20.01). *Students requiring testing accommodations must make arrangements with the DRO to take the exam; note that the DSO web site advises students to make reservations more than 3 days before the test date. We advise that students go ahead and make these reservations <u>at the beginning of the semester</u> since the test and exam dates are already fixed. The department does not have the facilities to accommodate special testing needs, which the DSO is set up to do.*

- **Inclusion Statement.** At NCSU, administrators, faculty, and staff are committed to the creation and maintenance of "inclusive learning" spaces, where you shall be treated with respect and dignity and where all individuals are provided equitable opportunity to participate, contribute, and succeed. In this course, all students are welcome regardless of race/ethnicity, gender identities, gender expressions, sexual orientation, socio-economic status, age, disabilities, religion, regional background, Veteran status, citizenship status, nationality and other diverse identities that we each bring to class. The success of an inclusive classroom relies on the participation, support, and understanding of you and your peers. We encourage you to speak up and share your views, but also understand that you are doing so in a learning environment in which we all are expected to engage respectfully and with regard to the dignity of all others.

- **Reporting Form.** A new departmental Anonymous Reporting Form is now online for use by students, staff, and faculty. Using it, you can communicate incidents and suggestions related to unprofessional behavior and departmental climate, including but not limited to cheating, bullying, and harassment.
- **Basic Needs Security.** Any student who faces challenges securing their food or housing or has other severe adverse experiences and believes this may affect their performance in the course is encouraged to notify the instructor if you are comfortable in doing so. Alternatively, you can contact the Division of Academic and Student Affairs to learn more about the Pack Essentials program.
- **Supporting Fellow Students in Distress:** As members of the NC State Wolfpack community, we each share a personal responsibility to express concern for one another and to ensure that this classroom and the campus as a whole remains a safe environment for learning. Occasionally, you may come across a fellow classmate whose personal behavior concerns or worries you. When this is the case, we would encourage you to report this behavior to the NC State Cares website. Although you can report anonymously, it is preferred that you share your contact information so they can follow-up with you personally.
- **ClassEval** is the end-of-semester survey for students to evaluate instruction of all university classes. Each semester students' responses are compiled into a ClassEval report for every instructor and class. Instructors use the evaluations to improve instruction and include them in their promotion and tenure dossiers, while department heads use them in annual reviews. Online class evaluations will be available for students to complete during the last two weeks of the semester for full semester courses and the last week of shorter sessions. Students will receive an email directing them to a website to complete class evaluations. Contact ClassEval Help Desk.
- **Religious/Cultural Observance.** Persons who have religious or cultural observances that coincide with this class should let the instructor know in writing (by e-mail for example) by January 31, 2025. I strongly encourage you to honor your cultural and religious holidays. However, if I do not hear from you by this date, I will assume that you do not have any attendance conflicts of this sort.
- **Title IX Statement.** Title IX makes it clear that violence and harassment based on sex or gender is a Civil Rights offense, subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race or national origin. If you or someone you know has been harassed or assaulted, you can find the appropriate resources here.